Hybrid guess-and-determine at (m, ℓ) = (3, 7): results

Study `research/dreg_hybrid_guess_20261024/`, pre-registered with four dated amendments. Class: stage diagnostic with a derived row; no S, no speed claim, no row beats rho. Host: shared 4-core 15 GB VM, not isolated.

Registered verdict

- **V-small.** On 10 of 10 unsatisfiable draws (8 old, 2 fresh) all 2^k branches refute at degree 4 for some $k^* \le 2$ (five draws $k^* = 1$, five $k^* = 2$).
- **Cheaper than the projected degree-5 closure** on every draw, at every exponent e and §5a weight g: 116 to 260 times at the e = 3 floor, 1,060 to 3,400 times at e = 4.1–4.25.
- **m = 4 row (derived): not improved at the exponent level.** Registered rule needs the S_lin row below +10.32 bits; with κ = 2 it is +58.27. Only the constant-k reading gives +8.43 bits (degree 4 floor +7.62, degree 5 +10.32). Constant versus growing k is unresolved; ℓ = 8 did not run.
- **Controls:** planted solutions survive (no solution-bearing branch refuted; the empty branches did). Placebo and A/A reproduce the records (columns, rank, outcome exact; word ops within 0.007 %). Strict reading: the registered product-count equality failed at 0.3–0.7 %, so the study is formally uninformative; amended reading (2 %) passes.

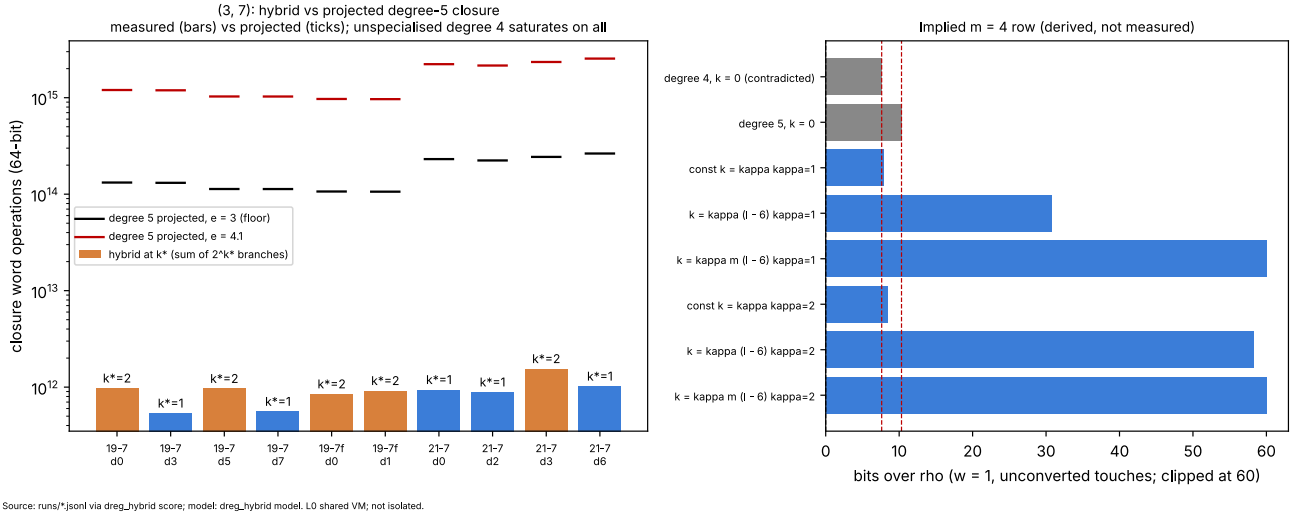


Figure 1: Left: hybrid word operations at k* (bars, measured) against the degree-5 closure projected from each draw's own unspecialised degree-4 cost (ticks, extrapolated). Right: implied m = 4 row, bits over rho at w = 1 (derived). Source: `plot.py` from `summary.json` and `runs/model-output.txt`.

Per draw

cell	draw	k*	k = 1	k = 2	ops at k*	with failed	RSS GB
(19,7)	0	2	fails	4 of 4	9.71e11	1.50e12	2.9
(19,7)	3	1	2 of 2	4 of 4 probe	5.34e11	5.34e11	3.1
(19,7)	5	2	fails	4 of 4	9.62e11	1.54e12	2.8
(19,7)	7	1	2 of 2	not run	5.56e11	5.56e11	6.0
(21,7)	0	1	2 of 2	not run	9.36e11	9.36e11	5.0
(21,7)	2	1	2 of 2	not run	8.84e11	8.84e11	4.9
(21,7)	3	2	fails	4 of 4	1.54e12	2.56e12	6.2
(21,7)	6	1	2 of 2	not run	1.02e12	1.02e12	5.1
fresh (19,7)	0	2	fails	4 of 4	8.45e11	1.10e12	3.0
fresh (19,7)	1	2	fails	4 of 4	9.14e11	1.44e12	2.7

Probe, (19,7) draw 3, k = 1, 2, 3: total 5.34e11, 8.11e11, 1.022e12 word ops (x1.52, x1.26 per guess; about 0.47 bit a guess; one draw).

Implied m = 4 row (derived, w = 1, bits over rho)

scenario	κ = 1	κ = 2
degree 4, k = 0 (floor)	+7.62	+7.62
degree 5, k = 0	+10.32	+10.32

S_const, k = κ	+7.95	+8.43
S_lin, k = κ(ℓ − 6)	+30.72	+58.27
S_linm, k = κ m (ℓ − 6)	+84.82	+84.82

§5a band: rho charged 0, 0.1, 1 unpriced operation per step moves S_const (κ = 2) to +8.43, +8.29, +7.43; the ranking against degree 5 is the same at every weight. Units are unconverted touches; log2 K of 7.4 to 8.4 bits would close the gap, K unmeasured.

Not run

(4, 7) infeasible (N ≈ 84, over 1 TB); E8 (3, 22, 8), fresh (21,7) draws and most probes did not run (lock contention).

Disclosures

- Runs outside the heavy-compute lock: the 21 branches in `runs/UNLOCKED.txt` (2 threads, at most 3.1 GB), 08:45 to 10:52 UTC; locked branches in that window overlapped them. No wall time here is comparable across runs; word-op counts are unaffected.
- No shared files changed. New: `examples/dreg_hybrid.rs`.